

Joel Teply

Senior / Staff / Principal Software Engineer — AI Systems · Distributed Systems · Full-Stack Engineering

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Profile

Computer engineer and full-stack software engineer since 1999 with deep experience building complete systems across backend, databases, web, cloud, native platforms, embedded hardware, computer vision, machine learning, distributed systems, and high-performance computing.

I have spent my career taking difficult systems from idea through architecture, implementation, integration, deployment, optimization, and production use. I am comfortable working wherever the problem requires: application code, APIs, data, infrastructure, models, runtimes, concurrency, GPU compute, hardware, or the product layer above them.

Co-founded and built Cambrian into a computer vision and AI company serving enterprise customers, while remaining a hands-on engineer and technical lead. Current work includes large-scale Rust AI systems, distributed execution, inference, continual learning, and model/runtime research.

Experience

Cambrian Tech — Co-founder, CTO, Principal Engineer · 2009–Present

Co-founded and built a computer vision and AI technology company serving enterprise customers across retail, home improvement, and automotive. Served initially as CEO and later CTO, leading product and technical strategy, engineering, hiring and team leadership, customer discovery, sales, partnerships, licensing, integration, and delivery while remaining deeply hands-on in architecture and implementation.

- Built commercial SDKs, applications, web platforms, backend services, and computer-vision systems used by enterprise customers and their

agency/development partners.

- Technology was licensed to major companies including Home Depot and Shaw Industries subsidiaries, reached millions of users, appeared across thousands of websites and in national television campaigns, and supported automotive spatial-computing work.
- Served as technical lead with customers from discovery and feasibility through architecture, integration, debugging, optimization, and delivery.
- Built production Python/TensorFlow ML services, APIs, S3-backed data pipelines, Docker CPU/GPU deployments, automated tests, model tooling, and elastically scaling AWS infrastructure.
- Designed and built full-stack product systems spanning browser applications, backend services, databases, native clients, SDKs, cloud infrastructure, and deployment.
- Architected real-time perception systems combining deep learning and classical computer vision with tracking, geometry, scene reconstruction, lighting/material analysis, and multi-frame fusion.
- Built high-performance C++ systems using concurrency, SIMD, GPU/GPGPU compute, scheduling/backpressure, careful memory movement, and platform-specific optimization.
- Built synthetic-data pipelines, model training/conversion/compression tooling, and ML systems spanning cloud inference and constrained on-device execution.
- Cambrian won the inaugural \$100,000 LaunchKC Grand Prize and was a top-five finalist at Atlanta Startup Battle 6.0.

Current AI Systems & Research — Cambrian · 2025–Present

Continuum — Sole architect of a large open-source Rust platform for persistent multimodal AI agents across commodity hardware.

- Built the system end to end across agent orchestration, tool execution, persistent memory, local inference, scheduling, evaluation, continual learning, distributed compute, peer-to-peer communication, and multimodal interaction.
- Designed resource-aware scheduling and memory management so multiple concurrent agents can share constrained model capacity.
- Built distributed movement of inference, training work, and learned adapters across heterogeneous machines, with provenance and reproducible evaluation artifacts.

- Extended open inference infrastructure and worked across Metal/CUDA execution, model placement, runtime behavior, and performance.
- Built evaluation-gated learning in which agent experience can generate training curricula and learned adaptations are retained only when measured performance improves.

Sentinel-AI — Transformer/model-efficiency research spanning attention-head plasticity, structured pruning, LoRA, mixed precision, quantization, evaluation, and reproducible model recipes.

AIRC — Rust peer-to-peer communication substrate for encrypted multi-agent messaging and file sharing across tabs, machines, and accounts.

github.com/CambrianTech/continuum

TripleBlind → Ideem — Software Engineer, Cybersecurity · 2023–2026

Built production software and enterprise SDK integrations for cryptographic authentication technology.

- Worked across client architecture, cross-platform SDKs, secure device identity/key handling, APIs, integration, debugging, and delivery.
- Contract role that later became full-time while Cambrian continued as an independent engineering effort.

EyeVerify / ZOLOZ — Software Engineer, Computer Vision & Biometrics · 2012–2016

One of the first two engineers turning biometric computer-vision research into a commercial authentication SDK on commodity phones; stayed through acquisition and integration.

- Built real-time C++ computer-vision systems for eye/iris tracking, segmentation, multi-frame fusion, image enhancement, and biometric verification.
- Developed GPU/threaded processing pipelines operating at camera frame rate on constrained hardware.
- Worked on presentation-attack and liveness detection and production SDK architecture.
- Named inventor on multiple patents arising from the work.

Propaganda3 — Software Engineer · 2010–2012

Built production mobile and web products for agency clients. Developed H&R Block's first tax app in roughly three months, including OCR/photo W-2 ingestion and a reusable engine supporting federal and state tax forms; the architecture was reused for Android. Also built other consumer fitness and media applications.

VML — Software Engineer · 2006–2010

Full-stack agency engineering for major brands across C#/.NET, CMS-backed web systems, frontend applications, databases/SQL, Java, web services, backend integrations, and customer-facing production systems.

AgileWise — Hardware & Software Engineer · 2005–2006

Designed PCB-level embedded hardware and a touchscreen computer/phone prototype; integrated display and RFID hardware, wrote display-driver/platform code, brought up Windows CE and Linux, and built supporting software and services.

Earlier Engineering · 1999–2006

Full-stack, backend, database, enterprise application, and infrastructure engineering through AllofE Solutions, Federal Home Loan Bank of Topeka, NovaStar Financial, VML contracting, and independent work.

- PHP, Perl, Java, LAMP/MySQL, ASP.NET/C#, SQL, B2B systems, web infrastructure, databases, and production application development.
- Repeatedly built systems end to end: relational schemas, SQL, custom ORMs, web servers, backend services, frontend applications, integrations, and deployment infrastructure.

Technical Breadth

Software / Systems: Rust, C, C++, Python, Java, C#, Go, TypeScript/JavaScript, PHP, Perl, Objective-C/Objective-C++, Swift, Kotlin, Lua, shell/Bash; concurrency, schedulers, memory systems, distributed systems, embedded/edge systems

Cloud / Backend / Data: AWS, S3, CloudFormation, Docker, serverless and scalable services, REST/GraphQL APIs, message-driven systems, automated testing, data/ML pipelines; SQL Server, MySQL, PostgreSQL, SQLite, NoSQL/Dgraph; schema design, custom ORMs and web servers

AI / ML: PyTorch, TensorFlow, LLMs, agents, RAG, LoRA, continual learning, evaluation, model routing, quantization, structured pruning, CNNs, metric learning, GANs, synthetic data, production inference

Performance / GPU / Native: Metal, CUDA, SIMD, OpenGL/GLSL, OpenGL ES GPGPU, WebGL, native camera/media pipelines, iOS, Android/JNI, embedded Linux, Windows CE, Unity, Unreal Engine

Web / Product: React, Node.js, TypeScript/JavaScript, HTML/CSS, browser graphics and ML, frontend/backend web systems, SDK design, enterprise integrations, product systems spanning web, mobile, cloud, and hardware

Education & Patents

B.S. Computer Engineering — University of Kansas

- US10964097B2 — Pattern Recognition Systems and Methods for Performing Segmentation on Surfaces and Objects; first-named inventor
- US9665784B2, US8437513B1, US8675925B2 — spoof detection / liveness